TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Dyeing of Blended Fabric, Dye Identification and Printing

Two fibres, one fabric, one level shade — plus the systematic identification of unknown dyes and first printing trials on cotton, polyester and blends.

WP-4102 · TEXTILE CHEMISTRY & COLORATION-III SESSIONAL · EXPERIMENT 2 · SPECIALIZATION — WET PROCESSING

1. Aim and theory

Aim: to dye a polyester/cotton blended fabric to a solid (union) shade by two routes — the **two-bath method** and the **one-bath two-step method** — to survey the dyeing logic of other common blends, to identify unknown dyes on fibre by a **systematic chemical test scheme**, and to carry out **printing trials** with reactive and pigment pastes on cotton and disperse paste on polyester and on the blend.

The blend problem. A blend is dyed correctly only when each component receives its own correct shade contribution. In a solid shade the two components must match each other in depth and hue when viewed together; if one component is paler, the fabric looks frosty or speckled (a fault called skitteriness when individual fibres differ). Polyester/cotton (P/C), the blend used here, is the hardest common case because its two fibres demand opposite conditions: polyester needs disperse dye at 130 °C in a mildly acidic bath, cotton needs reactive dye at 60 °C with salt and alkali. No single bath can be at two temperatures and two pH values at once — the whole technology of blend dyeing is the search for the least damaging compromise between the two sets of conditions.

Three further principles govern the work. First, **cross-staining**: each dye also deposits, to some degree, on the wrong fibre (disperse dye stains cotton; reactive dye can stain polyester slightly), and washing / clearing steps must remove that stain without removing the correctly fixed dye. Second, **reservation**: where a two-tone effect is wanted, one fibre is deliberately left undyed (reserved); the same chemistry, run unintentionally, is a fault. Third, **dye identification** reverses the dyeing process analytically: dyes are stripped or reacted in test reagents whose colour changes are characteristic of each class, allowing an unknown dyeing to be assigned to its class before a match or a strip is planned.

BLEND	DYE COMBINATION	STANDARD ROUTE AND KEY DIFFICULTY
Polyester / cotton	Disperse + reactive (or vat / direct)	Two-bath or one-bath two-step; temperature and pH conflict; clear disperse stain from cotton

Polyester / wool	Disperse + acid	Single bath near 100–106 °C with carrier or HT with wool protection; wool damaged at 130 °C
Wool / acrylic	Acid + basic	One bath, careful pH: two ionic dyes of opposite charge must not precipitate each other; retarder control
Viscose / cotton	Reactive or direct (same class)	One bath suffices chemically, but viscose exhausts faster / deeper — dye selection for equal uptake (tone-in-tone control)
Cotton / nylon	Reactive or direct + acid	Separate control of each component; nylon stains readily with cotton dyes

2. Apparatus and materials

HT dyeing machine with sealed beakers, laboratory dyeing bath, balance, beakers, thermometer, pH papers / meter, flat-bed screen or block and squeegee for printing trials, printing pastes and stock thickenings, drying oven, steamer or curing oven, multifibre adjacent fabric for staining checks. Fabrics: P/C 65/35 woven fabric (scoured and bleached), cotton fabric, polyester fabric; small wool, acrylic and viscose samples for the survey dyeings and identification unknowns. Unknown dyed samples (at least four, coded) supplied by the laboratory for Section 6.

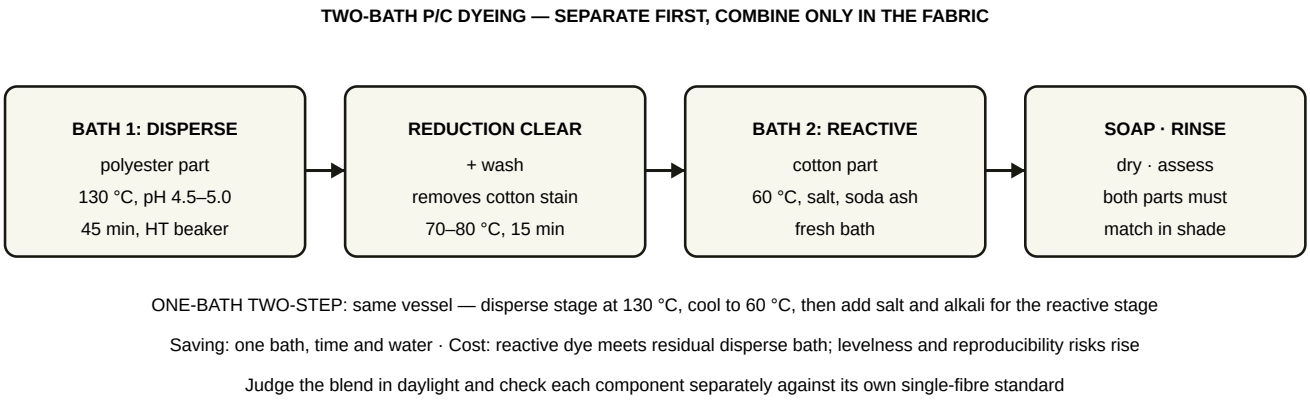


Fig. 1 — Two-bath sequence for polyester/cotton blend dyeing, with the one-bath two-step variant summarised below.

3. Blend composition and worked calculation

The fabric is P/C 65/35. For a 10 g sample the fibre masses, and the reference dye amounts for this sheet, are:

$$\text{Polyester in sample} = 10 \times 0.65 = 6.5\ \text{g} \quad \cdot \quad \text{Cotton in sample} = 10 \times 0.35 = 3.5\ \text{g}$$

ITEM	CALCULATION (ON TOTAL FABRIC WEIGHT)	AMOUNT
Disperse dye, 1.2% owf	10 × 1.2 ÷ 100	**0.12 g**
Reactive dye, 1.8% owf	10 × 1.8 ÷ 100	**0.18 g**
Disperse bath, LR 1:20	10 × 20	**200 mL**, pH 4.5–5.0
Reactive bath, LR 1:20	10 × 20	**200 mL**; salt and soda ash dosed in g/L exactly as in Experiment 1, Procedure A

Why the % owf basis matters: in this sheet both dyes are dosed on *total* fabric weight, which is the usual laboratory convention for blends and the basis of the pre-set recipes. Some dyehouses dose each dye on the weight of its own fibre component only (disperse on 6.5 g, reactive on 3.5 g here); the two conventions give very different dye weights, so the basis must always be written on the recipe card. An answer that gives a number without its basis is incomplete.

4. Procedure A — P/C blend, two-bath method

1. **Disperse stage (fresh bath 1).** Prepare 200 mL with disperse dye 0.12 g properly dispersed, dispersing agent 1 g/L and acetic acid to pH 4.5–5.0. Enter the pre-wetted 10 g P/C fabric at 50 °C in a sealed HT beaker, raise 2 °C/min to 130 °C, hold 45 min.
2. Cool below 80 °C, remove, rinse. The cotton component at this point carries loose disperse stain and looks nearly as coloured as the polyester — this is expected and is not yet the result.
3. **Reduction clearing.** Treat in caustic soda 2 g/L + hydrosulphite 2 g/L at 70–80 °C for 15 min, rinse and neutralise. Clearing strips the disperse stain from the cotton (leaving it white for its own dyeing) and clears the polyester surface. Inspect: the fabric should now show colour on the polyester component only.
4. **Reactive stage (fresh bath 2).** Prepare a second 200 mL bath with reactive dye 0.18 g and follow Experiment 1, Procedure A exactly: entry at 30 °C, salt in two portions, raise to 60 °C, soda ash in two portions, run 45 min.
5. Rinse, boil-soap (2 g/L detergent, 10 min), rinse and dry. The boil-soaping also removes any last disperse stain loosened from the cotton.
6. **Assessment.** Compare the blend against separate polyester-only and cotton-only standards of the target shade. View the fabric flat and also tease out yarns: both components must read as the same depth and hue. Record any frostiness, and which component is at fault.

5. Procedure B — P/C blend, one-bath two-step method

1. Prepare one 200 mL bath with the disperse dye 0.12 g, dispersing agent and acetic acid to pH 4.5–5.0. Carry out the HT stage exactly as in Procedure A, step 1 (130 °C, 45 min).
2. Without draining, cool the bath to 60 °C. Sample-check pH: the bath is still acidic and now contains exhausted and residual disperse dye and dispersing agent.
3. Add the reactive dye 0.18 g (pre-dissolved) and salt in two portions at 60 °C, allowing migration; then add soda ash in two portions to raise the pH into the fixation range and run 45 min at 60 °C. Alkali addition must be gradual: a sudden pH jump hydrolyses reactive dye and can disturb the disperse dyeing.
4. Rinse, boil-soap thoroughly (this soaping now does the work of both clearing the cotton stain and removing hydrolysed reactive dye), rinse and dry.
5. **Comparison.** Place Procedures A and B side by side with their records. Note differences in depth, levelness, handle, total time, water and chemical use. The one-bath route is the industrially attractive route (one bath filling, less water and time); its risks — reactive dye hydrolysis products and disperse residues in the same liquor, stain control resting on a single final soaping, and poorer reproducibility — are exactly what the comparison must expose.

Other blends — survey dyeings (demonstration / small scale)

Wool/acrylic: dye in one bath with an acid dye for wool and a basic dye for acrylic, with a retarder and careful pH control near 4.5–5. The two dye types are ionic with opposite charges; added carelessly they precipitate each other in the bath, so dyes are added separately and compatibility (a mixed-bath stability test) is checked first. **Viscose/cotton:** both components take the same reactive dyes in a single cotton-type bath, so no second bath is needed; the task is dye selection — choose dyes whose uptake ratio on viscose and cotton is near unity, otherwise the higher-exhausting viscose dyes deeper and the blend looks two-toned. Carry out one small dyeing of each pair as directed, record the bath conditions, and file the samples with the P/C results for comparison.

6. Dye identification on fibre — systematic scheme

Purpose and method. Given a dyed fabric of unknown class, assign the dye to its class using small cuttings (about 0.2 g per test) and the reagent sequence below. Work in order: burning / fibre identification first (the possible dye classes depend on the fibre), then solubility and stripping tests, then class-specific confirmations. Always run a known control sample beside an unknown when a colour change is subtle, and record every observation — including negative results — in a grid before concluding.

TEST (IN ORDER)	PROCEDURE	INTERPRETATION
1. Boiling water / DMF extraction	Boil cutting in water; separately warm in dimethylformamide (DMF)	Disperse dye on polyester extracts into DMF (fibre swells / dissolves in hot DMF) — first indication of disperse; most cotton dyes do not colour boiling water strongly
2. Alkaline hydrosulphite strip	Boil cutting in NaOH + hydrosulphite solution	Azo-based classes (direct, reactive, acid azo) decolourise; sulphur dyes decolourise and may re-darken in air; vat dyes change to leuco colour and restore on oxidation / airing; pigments largely resist stripping and flake with the binder
3. Re-oxidation check	Aire the stripped cutting from test 2	Colour returns → vat (or indigo-type) behaviour; colour stays destroyed → azo cleavage (direct / reactive / azoic chromophore destroyed)
4. Acid / alkali spot tests	Spots of dilute acetic acid and dilute ammonia / soda ash on separate cuttings	Marked reversible shade change suggests acid or basic dye character; basic dyes on acrylic brighten / shift characteristically; reactive dyeings change little (covalent, stable chromophore after cleavage tests considered)
5. Hypochlorite test	Warm dilute sodium hypochlorite on a cutting	Most organic dyes bleach; rate and residue help separate pigment (binder film resists briefly, colour rubs off mechanically) from fibre-bound dyes; use as supporting evidence only
6. Azoic confirmation	Cold dilute acid / solvent behaviour plus test 2 result	Azoic pigment resists aqueous stripping more than direct dye of similar shade, consistent with an insoluble pigment formed in the fibre rather than a soluble dye adsorbed on it

Recording rule: a class is identified only when at least two independent tests agree. A single colour change is evidence, not identification — several classes share azo chromophores and therefore share the hydrosulphite result, and they are separated by the fibre, the re-oxidation behaviour and the extraction tests taken together.

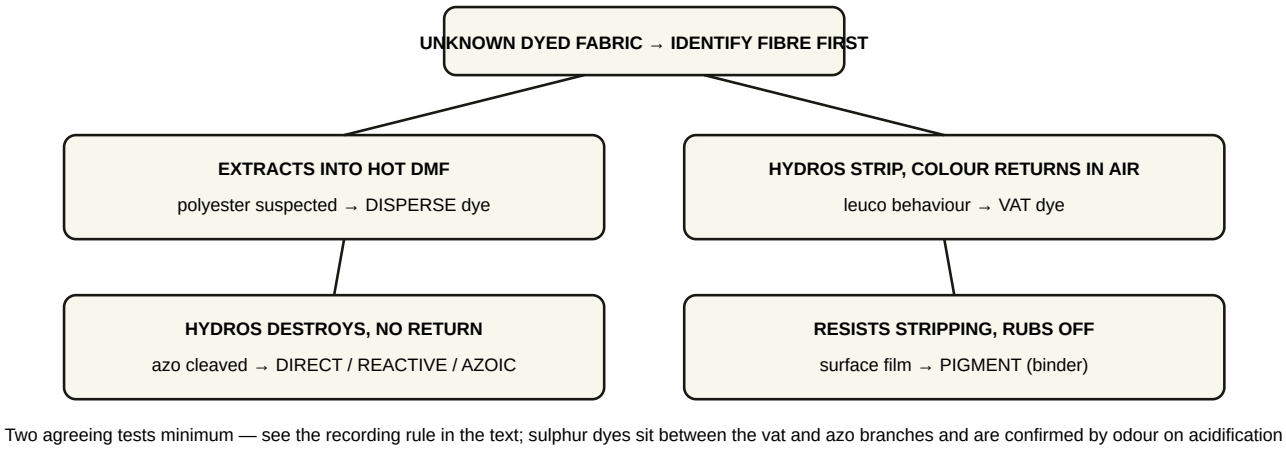


Fig. 2 — Dye-identification flowchart: first-branch decisions from the systematic test scheme.

7. Printing trials

Printing is localised coloration: a thickened paste is applied in a pattern, then the colour is fixed and the thickener washed out (pigment printing excepted — nothing is washed out). Each trial below uses a small screen or block, one squeegee pass unless stated, on a pinned, flat sample. Record paste recipe, number of passes, fixation and afterwash for every print — an unrecorded print cannot be reproduced.

7.1 Cotton — reactive print

Paste (per 100 g): reactive dye 3 g dissolved with urea 10 g and warm water, sodium alginate stock thickening about 80 g, soda ash (or sodium bicarbonate for steam fixation) 2 g, resist salt / mild oxidant 1 g, made up to 100 g. Print on cotton, dry, **steam 10 min** (saturated steam, about 102 °C) to fix, then wash off thoroughly — cold rinse, hot wash, boil-soap — and dry. The alginate thickener is chosen because it does not react with reactive dyes (starch thickenings do, and waste dye). Assess line sharpness, colour yield against the recipe depth, and wash-off staining of the white ground.

7.2 Cotton — pigment print

Paste (per 100 g): pigment dispersion 5 g, binder 15 g, emulsion or synthetic thickener to printing viscosity, softener as directed, to 100 g. Print, dry, **cure at 150 °C for 3 min**. No washing off. Compare handle and rubbing fastness with the reactive print: the pigment print sits on the surface, gives the sharpest lines on any ground colour, and its fastness is the binder film's fastness.

7.3 Polyester — disperse print

Paste (per 100 g): disperse dye 4 g dispersed in warm water, guar or alginate-based thickening to 100 g with a mild acid donor to hold pH 5–6. Print on polyester, dry carefully without migration, then fix by **HT steam at 170–175 °C for 7 min** (or thermosol / hot-air fixation as the laboratory directs), wash off

and reduction-clear lightly if directed. Note that fixation temperature, not paste composition, is the critical variable: an under-fixed disperse print washes down severely.

7.4 Blend — printing P/C

Print the P/C fabric with (a) the pigment paste of 7.2 and (b), where directed, a disperse + reactive combined approach. The pigment print colours both components in one operation and one fixation — which is why pigment printing dominates blend printing in industry. A disperse-only print colours only the polyester component, giving a deliberate two-tone print effect on the blend; file both prints and state, for each, which component carries the colour.

PRINT	FIXATION	AFTERTREATMENT / DEFINING PROPERTY
Reactive on cotton	Steam, about 102 °C, 10 min	Full wash-off + boil-soap; covalent fixation, soft handle
Pigment on cotton / blend	Cure 150 °C, 3 min	None; binder film, sharpest lines, stiffer handle
Disperse on polyester	HT steam 170–175 °C, 7 min	Wash-off ± light clearing; heat fixation is critical

8. Results and observations

ITEM	WHAT TO RECORD / ATTACH
Two-bath P/C dyeing	Sample; component match judgement; clearing effectiveness (cotton whiteness before reactive stage)
One-bath two-step dyeing	Sample beside two-bath sample; time / water comparison; levelness and reproducibility comment
Survey blends	Wool/acrylic and viscose/cotton samples with bath conditions and tone-match comment
Identification unknowns	Completed test grid for each coded unknown; concluded class with the two (or more) agreeing tests named
Prints	Four prints labelled with paste, fixation and wash-off; sharpness, colour yield, handle and ground-staining assessment

9. Precautions

- HT work: sealed beakers only, cool below 80 °C before opening — as in Experiment 1. Blend dyeings add a second hazard: a hot acidic disperse bath receiving alkali in the one-bath method must be cooled to 60 °C first and the alkali added gradually.
- DMF (identification tests): handle in a fume cupboard with gloves; it penetrates skin and carries dissolved substances with it. Hypochlorite: never mix with acid (chlorine gas).
- Printing pastes: keep screens and blocks wet between passes (dried paste blocks the mesh), clean thickener spills at once, and cure pigment prints with the oven timer set — scorched binder yellows the ground.
- Shade assessment of blends must be done dry and in the light cabinet; wet blends always look deeper and more level than they are.

10. Practice (viva preparation)

1. Why can polyester/cotton not be dyed well in a single bath held at one temperature and one pH throughout?
2. A 10 g P/C 65/35 sample is dyed with disperse 1.2% owf and reactive 1.8% owf, both on total weight. Give both dye weights and the mass of each fibre in the sample.
3. State two advantages and two risks of the one-bath two-step method against the two-bath method.
4. An unknown red on cotton is destroyed by alkaline hydrosulphite and does not return in air, while a second sample changes colour in the same bath and recovers on airing. Assign the behaviour of each and name the confirming evidence you still need.
5. Why is sodium alginate used for reactive printing, and why does pigment printing dominate the printing of blends?
6. What fault is skitteriness, and in which result in this experiment would you look for it?

Answer key

1. The components demand incompatible conditions: disperse dyeing of polyester needs about 130 °C at pH 4.5–5.0, reactive fixation on cotton needs about 60 °C with alkali after salt exhaustion. One fixed temperature / pH satisfies at most one component; blend methods therefore separate the stages in time (two-step) or in liquor (two-bath).
2. Disperse = 10 × 1.2 ÷ 100 = 0.12 g. Reactive = 10 × 1.8 ÷ 100 = 0.18 g. Polyester = 6.5 g; cotton = 3.5 g.
3. Advantages: one bath filling (saves water, energy and time) and a shorter total cycle. Risks: cross-staining control rests on the final soaping alone (no separate clearing bath), and reactive dye is exposed to residual disperse bath contents with gradual alkali addition — levelness and batch-to-batch reproducibility suffer.

4. First sample: azo chromophore cleaved irreversibly — consistent with a direct, reactive or azoic red; fibre (cotton) fits all three, so separation needs the extraction / stripping-resistance evidence of the scheme (for example azoic's resistance as an in-fibre pigment versus direct dye behaviour). Second sample: reversible leuco formation — vat dye behaviour; confirm by the characteristic leuco colour and full restoration on oxidation.
5. Alginate carries no free hydroxyl groups that compete for reactive dye in the way starch does, so colour yield is high and the thickener washes out cleanly. Pigment printing colours every fibre component at once through the binder film, with one paste and one curing step — no per-fibre dye selection or dual fixation is needed on a blend.
6. Skitteriness is a speckled, frosty appearance caused by individual fibres or one blend component dyeing to a visibly different depth from its neighbours. Look for it in the P/C dyeings (Procedures A and B) by comparing the two components against their single-fibre standards and by teasing out yarns.

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